Worksheet 4.9

Chemical formulae

Student:..... Class:.....

Covalent compounds

1. Complete the table by writing the chemical formula of the covalent compound formed from the listed elements.

Element 1	Element 2	Formula	Name of compound
Nitrogen	Chlorine		Nitrogen trichloride
Sulfur	Oxygen		Sulfur trioxide
Carbon	Sulfur		Carbon disulfide
Carbon	Oxygen		Carbon monoxide
Phosphorus	Oxygen		Diphosphorus pentoxide
Carbon	Fluorine		Carbon tetrafluoride
Oxygen	Bromine		Oxygen dibromide
Nitrogen	Oxygen		Dinitrogen tetroxide

Ionic compounds

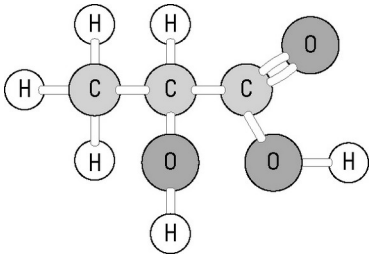
2. Use the table of valencies in your textbook to complete the table by writing the chemical formula of the ionic compound formed from the listed elements. Name each compound.

Positive ion	Negative ion	Formula	Name of compound
Silver	Iodide		
Lead	Sulfide		
Calcium	Nitride		
Aluminium	Sulfide		
Barium	Nitrate		
Copper (II)	Sulfate		
Sodium	Phosphate		
Zinc	Hydrogen carbonate		

Structural formulae

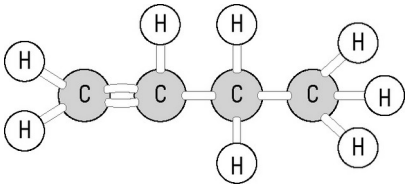
3. The structural formulae for various common compounds are shown. Write the molecular formula for each compound.

(a) Lactic acid



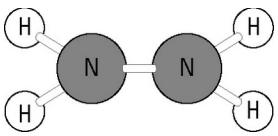
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(b) Butene



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(c) Hydrazine



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